

3.3.1 Atoms, elements and compounds

Students should have knowledge and understanding of the following content.

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Outcome 1			
All substances are made of atoms. An atom is the smallest part of an element that can exist.		5.1.1.1	
A substance that is made of only one sort of atom is called an element. There are about 100 different elements. Elements are shown in the periodic table. Metals are towards the left and the bottom of the periodic table and non-metals towards the right and the top of the periodic table. Students should know that most of the elements are metals.		5.1.1.1 and 5.1.2.3	4.5.1.1 and 4.5.1.2
Elements in the same group of the periodic table have similar chemical properties.	Knowledge of groups is limited to Group 1 as reactive metals and Group 7 as reactive non-metals.	5.1.2.1	4.5.1.1
Outcome 2			
When elements react, their atoms join with other atoms to form compounds.		5.1.1.1	4.5.2.1
Some compounds are made from metals combined with non-metals, for example sodium chloride and magnesium oxide. Students should be able to recognise simple compounds from their names, eg sodium chloride, magnesium oxide, carbon dioxide. Some compounds are made from only non-metals, for example carbon dioxide.		5.2.1.2	4.6.2.1

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Chemical reactions can be represented by word equations. Students should be able to write word equations for reactions of metals and non-metals, reactions of non-metals to produce oxides, and the other chemical reactions in this specification.		5.1.1.1	4.5.2.1